

Technology Selection and Rationale

DIRECTIONAL · AS AT 3 SEPTEMBER 2026

Directional. First-pass selections for V3, not settled decisions. This view answers which technologies the platform uses and why: the problem each one solves, the features actually relied on, and what was rejected. It deliberately draws no relationships — **LB-V3-L2-01** owns how the platform is constructed. The argument for each selection lives in its ADR; where no V3 ADR exists the card says so, and that gap is a to-do rather than a defect.

Vue 3 + PrimeVue

PROVEN IN LAB

Experience channel – role-shaped workspaces

SOLVES

Several role-shaped workspaces must look and behave as one product without one design system per team.

RELIES ON

Composition API, Pinia stores as client-side read models, PrimeVue v4 theming preset, vue-i18n

INSTEAD OF

React plus a component library; a hand-rolled design system

DECISION

no V3 ADR yet

USED BY

App Shell + MFE; Operator + tenant admin; Driver mobile

Module Federation

PROVEN IN LAB

Runtime composition of micro-frontends

SOLVES

A team must ship a workspace feature without rebuilding or redeploying the shell.

RELIES ON

Runtime remote loading, shared dependency singletons, per-remote versioning

INSTEAD OF

Build-time monorepo bundling; iframes

DECISION

no V3 ADR yet

USED BY

App Shell + MFE; Web delivery (shell + MFE registry)

WorkOS / enterprise IdP

PROPOSED

Identity federation – authentication only

SOLVES

Enterprise customers bring their own single sign-on, and Linebooker must not hold their passwords.

RELIES ON

SSO and directory sync, OIDC and SAML connections, hosted login

INSTEAD OF

Self-hosted Keycloak; bespoke SSO per tenant

DECISION

no V3 ADR yet; commercial selection outstanding

USED BY

Identity federation; Identity + policy broker

OpenFGA

PROPOSED

Relationship authorisation – who may act on what

SOLVES

Authority is relational, not a role name: a user may act on one organisation, one depot or one load, and V2 could not express that.

RELIES ON

Zanzibar-style relationship tuples, an authorisation model per tenant, batch check and list-objects queries

INSTEAD OF

Role enums in the database; CASL rules duplicated in the frontend; per-service bespoke checks

DECISION

no V3 ADR yet; self-hosted or hosted not decided

USED BY

Identity + policy broker; every domain service

Go

PROVEN IN LAB

Domain service implementation language

SOLVES

Services must start fast and run cheaply as many replicas in every cell.

RELIES ON

Static binaries, first-class NATS and Temporal SDKs, goroutine concurrency, hexagonal layout

INSTEAD OF

Java with Spring; Node.js

DECISION

no V3 ADR yet

USED BY

Every tile in the domain service plane; service replicas per cell

NATS Core

PROVEN IN LAB

Service-to-service messaging

SOLVES

Services must call each other and announce change without point-to-point HTTP wiring or a service mesh.

RELIES ON

Subject taxonomy, request/reply, queue groups, accounts as hard isolation boundaries

INSTEAD OF

gRPC with a service mesh; REST between services

DECISION

no V3 ADR yet

USED BY

NATS Core + JetStream; NATS provisioning; NATS logical HA per cell

NATS JetStream

PROVEN IN LAB

Durable fact and event history

SOLVES

Some state must be reconstructable and auditable, not only current.

RELIES ON

LimitsPolicy retention so history can be replayed, durable consumers, one domain per cell

INSTEAD OF

Kafka; an outbox table in every service database

DECISION

no V3 ADR yet

USED BY

NATS Core + JetStream; JetStream ZA, AU and BW

NATS KV

PROVEN IN LAB

Derived read model and fast lookup

SOLVES

Hot reads must not reach Postgres, and clients need to watch a value for change.

RELIES ON

Watch, context-scoped keys, write-through from the handler that also writes Postgres

INSTEAD OF

Redis; caches held inside each service process

DECISION

no V3 ADR yet

USED BY

Projections + NATS KV

Temporal

PROVEN IN LAB

Long-running process coordination

SOLVES

Tender expiry, execution, proof, claims and settlement run for days and span many services.

RELIES ON

Durable timers, automatic retry, workflow state as truth, deterministic replay

INSTEAD OF

Saga logic written into each service; cron with an outbox

DECISION

ADR-047, lab scope only; no V3 ADR yet for V3

USED BY

Temporal workflows; Temporal runtime per cell

PostgreSQL

CHOSEN

Transactional source of truth

SOLVES

Business rules need ACID transactions and a queryable current state owned by one service.

RELIES ON

ACID transactions, one database and one role per service, JSONB, shared instance by default

INSTEAD OF

A dedicated instance per service; one shared schema for all services

DECISION

ADR-052 and ADR-053; ADR-053 is V3 scope

USED BY

Service-owned Postgres; Service data per cell

S3-compatible object storage

PROPOSED

Evidence document repository

SOLVES

Proof of delivery, licences and certificates are large files that must not travel through messaging.

RELIES ON

The S3 API contract, presigned URLs, a bucket per cell, lifecycle rules

INSTEAD OF

NATS Object Store as the default; document blobs held in Postgres

DECISION

ADR-048 chose NATS Object Store for the lab; V3 differs and has no V3 ADR yet

USED BY

Document store adapter; Evidence store per cell

OpenTelemetry / OTLP

PROVEN IN LAB

Telemetry contract for logs, metrics and traces

SOLVES

Cells run independently, so telemetry must be collected the same way everywhere and stay in region.

RELIES ON

The OTLP wire format, vendor-neutral SDKs, masking at the collector

INSTEAD OF

A vendor agent embedded in every service

DECISION

no V3 ADR yet

USED BY

Observe + audit; Resident operations per cell

ULID

PROVEN IN LAB

Entity identity format

SOLVES

Identifiers appear in NATS subjects and KV keys, so they must be subject-safe and time-sortable.

RELIES ON

26 Crockford base32 characters, lexicographic time ordering, minted in application code

INSTEAD OF

UUIDv4; identifiers generated by the database

DECISION

ADR-051, lab scope only; no V3 ADR yet for V3

USED BY

organizations-service identity; every subject and KV key token

Container platform

OPEN

Where services actually run

SOLVES

Services must be packaged and scheduled the same way in every region cell, with room to run a sovereign cell separately.

RELIES ON

Nothing yet; the lab runs Docker Compose on one host

INSTEAD OF

V2 ran the API directly on Compute Engine virtual machines with no orchestrator

DECISION

no V3 ADR yet; Kubernetes, a managed container runtime and plain VMs are all still open

USED BY

Every deployed element in LB-V3-L2-01 sheet 2

Cloud provider

OPEN

The infrastructure the cells sit on

SOLVES

Regional cells and data residency rules must be satisfied by real regions in a real provider.

RELIES ON

Nothing yet; only the S3 and OTLP API contracts are assumed portable

INSTEAD OF

V2 is wholly on Google Cloud in one project

DECISION

no V3 ADR yet; a deliberate open decision, not an oversight

USED BY

Every regional cell; object storage; managed data services

Build and release pipeline

OPEN

How code reaches a cell

SOLVES

Independently deployed micro-frontends and services need independent, repeatable release paths.

RELIES ON

Nothing yet; the lab builds with Vite and Go toolchains directly

INSTEAD OF

V2 uses Google Cloud Build with fifteen pipelines, plus GitHub Actions for notifications

DECISION

no V3 ADR yet

USED BY

App Shell + MFE registry; every service

SCOPE

This is an inventory, not a topology. It answers what and why. LB-V3-L2-01 answers how the parts fit together; the two sit at L2 because they ask different questions.

STATUS

One decision is accepted for V3. Only ADR-053 is V3-scoped. Thirteen of the sixteen cards carry no V3 ADR at all, and two cite a lab-scoped ADR that does not accept the technology for V3.

ADR

The ADR is the record, not this sheet. Each card gives one line of reasoning and a pointer. Do not restate an argument here; it will drift and the diagram will win.

LAB

Proven in lab is not chosen. The Dictionary POC exercised NATS, JetStream, KV, Temporal, Go and the frontend stack at demo scale, under one operator, with no regulated data.

GAP

Read the amber text as a work list. Each "no V3 ADR yet" marks a selection the platform already leans on without a written decision behind it.

STORES

Four stores, four jobs. Postgres is transactional truth, JetStream is durable fact history, KV is a derived read model, and object storage holds evidence documents. They are not interchangeable.

VENDOR

No cloud vendor is selected. S3-compatible names an API contract that several vendors implement. Compute, network and managed-service vendors remain open decisions.

DIVERGE

Two lab decisions do not carry into V3 as written. ADR-048 chose NATS Object Store for lab documents; V3 proposes an S3-compatible repository instead. ADR-051 is scoped to one lab service.

DERIVED

Versions, sizing and configuration are out of scope. Pinned versions, high-availability topology, runbooks, subjects and schemas are L3 and L4 decisions.

REVIEW

A directional document expires. Re-date it whenever a card changes. A position with no date cannot be trusted a year from now.

How to read a card.

The left edge colours the plane the technology serves, using the same colour meanings as LB-V3-L2-01. The chip states the selection, not the document: **Chosen** is accepted for V3 with an ADR of record, **Proposed** is a directional position with no V3 ADR yet, and **Proven in lab** was exercised in the Dictionary POC but is not yet accepted for V3. **Used by** names the LB-V3-L2-01 elements that depend on the selection. No cloud vendor is listed: object storage is an API contract here, not a product choice.

KEY

	Blue edge — experience and entry plane		Purple edge — domain service plane
CHOSEN	Accepted for V3, with an ADR of record	PROPOSED	Directional position; no V3 ADR yet
USED BY	The LB-V3-L2-01 elements that depend on this selection	realises	The role the technology fills, stated before the product

	Cyan edge — platform foundations		Green edge — cross-cutting platform concern
PROVEN IN LAB	Exercised in the Dictionary POC; not yet accepted for V3	no V3 ADR yet	A selection with no written V3 decision behind it
no lines	An inventory view; relationships are drawn in LB-V3-L2-01	DIRECTIONAL	A first-pass position, dated on the sheet, not a settled decision

ACRONYMS

ACID Atomic, consistent, isolated, durable	POC Proof of concept
ADR Architecture decision record	S3 The object-storage API; S3-compatible means any vendor implementing it
API Application programming interface	SAML Security Assertion Markup Language
HA High availability	SDK Software development kit
IdP Identity provider	SSO Single sign-on
JSONB PostgreSQL binary JSON column type	ULID Universally unique lexicographically sortable identifier
KV Key/value store (NATS KV)	UUID Universally unique identifier
MFE Micro-frontend	ZA / AU / BW South Africa / Australia / Botswana (ISO country codes)
NATS The messaging system; a product name, not an abbreviation	
OIDC OpenID Connect	
OTLP OpenTelemetry protocol	

External Services and Integrations

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Directional. Third-party services the platform depends on, carried forward from V2 unless stated. Sheet 1 covers technologies Linebooker builds on; this sheet covers services Linebooker buys or integrates with, where the vendor owns the roadmap and the outage. Every card here is inherited operational reality, not a fresh V3 selection — none has been re-tendered, and the final card records what is deliberately not carried forward.

Acumatica PROPOSED Accounting and finance system of record SOLVESInvoices, debtors and customer master data live in a finance system Linebooker does not own. RELIES ONREST contract API, OAuth token exchange, customer and vendor entity sync INSTEAD OFBuilding invoicing inside the platform; the live Google Sheet V2 still reads debtors from DECISIONno V3 ADR yet; carried from V2, replacement not assessed USED BYFinancial architecture; billing and settlement	monday.com PROPOSED Commercial pipeline and account boards SOLVESSales and account teams work customers and transporters outside the operational platform. RELIES ONBoard and item API, item identifiers stored against transporters INSTEAD OFA customer-relationship module inside Linebooker DECISIONno V3 ADR yet; carried from V2, a candidate to retire USED BYOperator workspace; participant onboarding	Meta WhatsApp Cloud API PROPOSED Conversational channel to drivers and transporters SOLVESTransporters and drivers work from a phone and will not install or log in to a portal for every update. RELIES ONCloud API graph endpoints, message templates, inbound webhooks INSTEAD OFNative mobile push alone; SMS for every interaction DECISIONno V3 ADR yet; carried from V2, load-bearing in V2 USED BYDriver mobile; notification fan-out	SendGrid PROPOSED Transactional email delivery SOLVESThe platform must send templated, deliverable email at volume and prove it was sent. RELIES ONTemplate API, event webhooks, domain authentication INSTEAD OFSelf-hosted SMTP; the Spring mail fallback V2 still carries DECISIONno V3 ADR yet; carried from V2, vendor-swappable USED BYNotification fan-out; document distribution
Twilio Verify PROPOSED Second factor over SMS and voice SOLVESUsers without an enterprise identity provider still need a second factor. RELIES ONVerify service, one-time passcode issue and check INSTEAD OFRolling one-time passcodes in-house; authenticator app only DECISIONno V3 ADR yet; carried from V2; overlaps WorkOS multi-factor, one should go USED BYIdentity federation; Identity + policy broker	Peach Payments PROPOSED Card acceptance in South Africa SOLVESCustomers must pay by card without card data touching Linebooker systems. RELIES ONHosted checkout, entity identifiers per brand, status callbacks INSTEAD OFHolding card data; a global-only processor with no local acquiring DECISIONno V3 ADR yet; carried from V2; region-specific, not portable to AU USED BYFinancial architecture; payment capture	Ozow PROPOSED Instant bank transfer in South Africa SOLVESMany South African customers pay by electronic funds transfer, not card. RELIES ONPayment request API, transaction status polling INSTEAD OFManual proof-of-payment reconciliation DECISIONno V3 ADR yet; carried from V2; region-specific, not portable to AU USED BYFinancial architecture; payment capture	Telematics broker PROPOSED One tracking contract over many vendors SOLVESAround thirty-eight telematics vendors each expose a different feed, and a transporter may use any of them. RELIES ONPer-transporter vendor credentials, normalised position and event stream INSTEAD OFA direct integration per vendor inside each domain service DECISIONno V3 ADR yet; carried from V2 as Linetracker; V3 shape not designed USED BYTracking and visibility; load execution
Google Maps Platform PROPOSED Geocoding, distance and routing SOLVESAddresses must resolve to coordinates and legs must be priced and timed by real road distance. RELIES ONGeocoding, distance matrix and directions APIs INSTEAD OFOpen routing engines self-hosted; straight-line distance DECISIONno V3 ADR yet; carried from V2; cost scales with volume USED BYPricing; route planning; operating areas	MapLibre GL + MapTiler PROPOSED Map rendering in the browser SOLVESOperators need interactive maps of operating areas and live positions without a per-view licence cost. RELIES ONVector tile serving, self-hosted boundary tiles, open rendering library INSTEAD OFRendering every map through the Google Maps JavaScript API DECISIONno V3 ADR yet; carried from V2; splits map cost from map data USED BYOperator workspace; operating areas	Google Document AI PROPOSED Extraction from scanned evidence SOLVESProof of delivery arrives as a photograph or scan and must become structured data. RELIES ONA trained processor, batch and online extraction INSTEAD OFManual capture; a generic optical character reader with no document model DECISIONno V3 ADR yet; carried from V2; the strongest single cloud lock-in USED BYEvidence store; proof of delivery workflow	BigQuery + Looker Studio PROPOSED Analytics warehouse and reporting SOLVESOperational stores must not serve analytics, and customers ask for their own reports. RELIES ONWarehouse dataset, signed embedded reports INSTEAD OFReporting straight off the transactional database DECISIONno V3 ADR yet; carried from V2; a projection target, never a source of truth USED BYFinancial architecture; customer reporting
Customer file drops PROPOSED Contracted file exchange with large shippers SOLVESSome enterprise customers accept documents only over their own secure file transfer endpoint. RELIES ONSecure file transfer client, per-customer credentials and folder rules INSTEAD OFEmail attachments; asking the customer to use the portal DECISIONno V3 ADR yet; carried from V2 as a standalone middleware service USED BYExternal integration; document distribution	Voucher rewards PROPOSED Driver and transporter incentives SOLVESDrivers earn rewards that must be redeemed for something they actually want. RELIES ONRetailer voucher denominations and redemption INSTEAD OFCash incentives; a points scheme with no redemption path DECISIONno V3 ADR yet; carried from V2; single retailer, region-specific USED BYDriver mobile; rewards	reCAPTCHA PROPOSED Bot protection on public forms SOLVESPublic sign-up and enquiry forms are scripted against. RELIES ONServer-side token verification INSTEAD OFRate limiting alone DECISIONno V3 ADR yet; carried from V2; a candidate to fold into the edge USED BYPublic web entry; participant onboarding	Not carried into V3 NOT CARRIED V2 technologies deliberately left behind SOLVESA rewrite is the moment to stop paying for scaffolding no one uses. RELIES ONNothing; listed so the omission is visible rather than accidental INSTEAD OFJHipster scaffolding, Spring Boot 2.1, MySQL, RabbitMQ, Redis cache, Elasticsearch, Logstash, Dropwizard metrics, React 17 + Redux + Material UI, CASL, jOOQ, Liquibase DECISIONDirectional: each of these has a V3 replacement on sheet 1 or is simply dropped USED BYNothing; this card exists to close the inventory

How to read a card.

The amber left edge marks an external or separately governed service — the same colour meaning as LB-V3-L2-01. Carried from V2 means the dependency exists in the running platform today and has not been reassessed for V3; it is a statement of fact, not an endorsement. Used by names the LB-V3-L2-01 elements that would depend on the service.

KEY

	Amber edge — external or separately governed service		Grey edge — not a selection; a record of what is dropped	PROPOSED	Directional position; no V3 ADR yet	NOT CARRIED	Present in V2 and deliberately left behind
carried from V2	Runs in the platform today; not reassessed for V3	no V3 ADR yet	A dependency with no written V3 decision behind it	USED BY	The LB-V3-L2-01 elements that depend on the service	realises	The role the service fills, stated before the vendor
no lines	An inventory view; relationships are drawn in LB-V3-L2-01	DIRECTIONAL	A first-pass position, dated on the sheet, not a settled decision	no vendors	No endpoint, key or account identifier appears on this sheet	V2	The Linebooker platform running in production today

ACRONYMS

ADR	Architecture decision record	OCR	Optical character recognition
API	Application programming interface	POD	Proof of delivery
AU	Australia (ISO country code)	REST	The HTTP resource API style
BI	Business intelligence	SFTP	Secure file transfer protocol
CRM	Customer relationship management	SMS	Short message service
EFT	Electronic funds transfer	V2 / V3	The current and proposed Linebooker platform versions
ERP	Enterprise resource planning	ZA	South Africa (ISO country code)
IdP	Identity provider		
OAuth	The delegated authorisation standard		